National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computer Science

Department of Computing

CS220: Database Systems

Class: BSAI – 2A

Lab 12: Normalization

CLO-2: Formulate SQL queries to retrieve information from a relational database.

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Time: 10:00 - 01:00

Introduction

DBMS Normalization is a systematic approach to **decompose (break down) tables** to eliminate data redundancy(repetition) and undesirable characteristics like Insertion anomaly in DBMS, Update anomaly in DBMS, and Delete anomaly in DBMS.

Tools/Software Requirement

- MS Word

Lab Tasks:

Normalize the following to UNF, 1NF, 2NF and 3NF.

Task 1:

HEALTH HISTORY REPORT

PET ID	PET NAME	PET TYPE	PET AGE	OWNER	VISIT DATE	PROCEDURE
246	ROVER	DOG	12	SAM COOK	JAN 13/2002	01 - RABIES VACCINATION
					MAR 27/2002	10 - EXAMINE and TREAT WOUND
					APR 02/2002	05 - HEART WORM TEST
298	SPOT	DOG	2	TERRY KIM	JAN 21/2002	08 - TETANUS VACCINATION
					MAR 10/2002	05 - HEART WORM TEST
341	MORRIS	CAT	4	SAM COOK	JAN 23/2001	01 - RABIES VACCINATION
					JAN 13/2002	01 - RABIES VACCINATION
519	TWEEDY	BIRD	2	TERRY KIM	APR 30/2002	20 - ANNUAL CHECK UP
					APR 30/2002	12 - EYE WASH

UNF (Unnormalized Form):

In UNF, data may contain repeating groups.UNF is a table that may contain repeating groups and multi-valued attributes(same primary key in multiple rows with different corresponding results in other attributes.

The given table is already in UNF.

PET_ID	PET_NAME	PET_TYPE	PET_AGE	OWNER	VISIT_DATE	PROCEDURE
246	Rover	Dog	12	Sam Cook	{Jan 13 2002, Mar 27 2002, Apr 02 2002}	{Rabies Vaccination, Examine & Treat Wound, Heartworm Test}
298	Spot	Dog	2	Terry Kim	{Jan 21 2002, Mar 10 2002}	{Tetanus Vaccination, Heartworm Test}
341	Morris	Cat	4	Sam Cook	{Jan 23 2001, Jan 13 2002}	{Rabies Vaccination, Rabies Vaccination}
519	Tweedy	Bird	2	Terry Kim	{Apr 30 2002, Apr 30 2002}	{Annual Checkup, Eye Wash}

1NF (First Normal Form):

It removes repeating groups and make all values atomic. There wont be more than one component in an attribute e.g. 3 visit dates for 1st pet_id , each will now be written in separate rows under same pet_id.

PET_ID	PET_NAME	PET_TYPE	PET_AGE	OWNER	VISIT_DATE	PROCEDURE
246	Rover	Dog	12	Sam Cook	Jan 13 2002	Rabies Vaccination
246	Rover	Dog	12	Sam Cook	Mar 27 2002	Examine and Treat Wound
246	Rover	Dog	12	Sam Cook	Apr 02 2002	Heart Worm Test
298	Spot	Dog	2	Terry Kim	Jan 21 2002	Tetanus Vaccination
298	Spot	Dog	2	Terry Kim	Mar 10 2002	Heart Worm Test
341	Morris	Cat	4	Sam Cook	Jan 23 2001	Rabies Vaccination
341	Morris	Cat	4	Sam Cook	Jan 13 2002	Rabies Vaccination
519	Tweedy	Bird	2	Terry Kim	Apr 30 2002	Annual Check Up
519	Tweedy	Bird	2	Terry Kim	Apr 30 2002	Eye Wash

2NF (Second Normal Form):

- Must be in 1NF
- No **partial dependency** on composite key
- PET_NAME, PET_TYPE, PET_AGE, OWNER
  depend only on **PET_ID**, not on the full key.
- Therefore we create 2 separate tables with same primary key.

PET Table:

PET_ID	PET_NAME	PET_TYPE	PET_AGE	OWNER
246	Rover	Dog	12	Sam Cook
298	Spot	Dog	2	Terry Kim
341	Morris	Cat	4	Sam Cook
519	Tweedy	Bird	2	Terry Kim

VISIT Table:

PET_ID	VISIT_DATE	PROCEDURE
246	Jan 13 2002	Rabies Vaccination
246	Mar 27 2002	Examine and Treat Wound
246	Apr 02 2002	Heart Worm Test
298	Jan 21 2002	Tetanus Vaccination
298	Mar 10 2002	Heart Worm Test
341	Jan 23 2001	Rabies Vaccination
341	Jan 13 2002	Rabies Vaccination
519	Apr 30 2002	Annual Check Up
519	Apr 30 2002	Eye Wash

3NF (Third Normal Form):

A relation is in Third Normal Form (3NF) if:

1. It is in Second Normal Form (2NF)
2. There is **no transitive dependency**
   (i.e., non-key attributes do not depend on other non-key attributes)
3. Here I created separate Ids for independent attributes like pet_id, owner_id etc.

PET Table:

PET_ID	PET_NAME	PET_TYPE	PET_AGE	OWNER_ID
246	Rover	Dog	12	1
298	Spot	Dog	2	2
341	Morris	Cat	4	1
519	Tweedy	Bird	2	2

OWNER Table:

OWNER_ID	OWNER_NAME
1	Sam Cook
2	Terry Kim

PROCEDURE Table:

PROC_ID	PROCEDURE_NAME
01	Rabies Vaccination
05	Heart Worm Test
08	Tetanus Vaccination
10	Examine and Treat Wound
12	Eye Wash
20	Annual Check Up

VISIT Table:

PET_ID	VISIT_DATE	PROC_ID
246	Jan 13 2002	01
246	Mar 27 2002	10
246	Apr 02 2002	05
298	Jan 21 2002	08
298	Mar 10 2002	05
341	Jan 23 2001	01
341	Jan 13 2002	01
519	Apr 30 2002	20
519	Apr 30 2002	12

Task 2:

INVOICE

HILLTOP ANIMAL HOSPITAL DATE: JAN 13/2002
INVOICE # 987

MR. RICHARD COOK
123 THIS STREET
MY CITY, ONTARIO
Z5Z 6G6

PET	PROCEDURE	AMOUNT
ROVER	RABIES VACCINATION	30.00
MORRIS	RABIES VACCINATION	24.00
	TOTAL	54.00
	TAX (8%)	4.32
	AMOUNT OWING	58.32

UNF (Un-Normalized Form):

Identified attributes namely Invoice_no, date, customer_name, address, pet, procedure, amount, total, tax, amount owing. Then I arranged them into a table form without addressing any normal forms which resulted in a UNF table. Contains repeating groups (PET, PROCEDURE, AMOUNT) so not normalized

INVOICE_NO	DATE	CUSTOMER_NAME	ADDRESS	PET	PROCEDURE	AMOUNT	TOTAL	TAX	AMOUNT_OWING
987	Jan 13 2002	Richard Cook	123 This Street	{Rover, Morris}	{Rabies Vaccination, Rabies Vaccination}	{30.00, 24.00 }	54.00	4.32	58.32

1NF (First Normal Form):

All attributes are atomic (no repeating groups)

INVOICE_NO	DATE	CUSTOMER_NAME	ADDRESS	PET	PROCEDURE	AMOUNT
987	Jan 13 2002	Richard Cook	123 This Street	Rover	Rabies Vaccination	30.00
987	Jan 13 2002	Richard Cook	123 This Street	Morris	Rabies Vaccination	24.00

2NF (Second Normal Form):

INVOICE Table:

INVOICE_NO	DATE	CUSTOMER_NAME	ADDRESS	TOTAL	TAX	AMOUNT_OWING
987	Jan 13 2002	Richard Cook	123 This Street, Ontario, Z5Z6G6	54.00	4.32	58.32

INVOICE-DETAILS Table:

INVOICE_NO	PET	PROCEDURE	AMOUNT
987	Rover	Rabies Vaccination	30.00
987	Morris	Rabies Vaccination	24.00

3NF (Third Normal Form):

The relation is now in Third Normal Form (3NF) because:

- All repeating groups are removed
- No partial dependency exists
- No transitive dependency exists
- Each non-key attribute depends only on the primary key
- The tables also satisfy conditions for 1NF and 2NF

CUSTOMER_ID	CUSTOMER_NAME	ADDRESS
1	Richard Cook	123 This Street

Invoice No	Date	Customer ID	Total	Tax	Amount Owing
987	Jan 13 2002	1	54.00	4.32	58.32

Pet ID	Pet Name
1	Rover
2	Morris

Proc ID	Procedure Name
01	Rabies Vaccination

Invoice No	Pet ID	Proc ID	Amount
987	1	01	30.00
987	2	01	24.00